Milestone 7: Claims Validation, Attribution and RAF

How closures tie to claims, whether a reward can be credited, and the separate risk-adjustment arm

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RESEARCH FRAMEWORK. Lift is research-generated or taken from published trials; no InComm lift, financial or issuance data is used. Dollar values are research ranges tagged by provenance.

1. Closure referential-integrity check (not an independent validation)

Referential-integrity check on InComm's export (claim ids MOCK-CLM-*). Closure is derived from the same claim, so member, CPT, date and window agree by construction; 100% shows internal consistency, not clinical validity or production data quality. Gaps sharing a claim map to the same HEDIS measure (e.g. DIAB_A1C_TEST and DIAB_BLOOD_SUGAR both -> CDC-A1C), so measure value is de-duplicated across them.

Care Gap Code	Closed Gaps	Claim Found	Same Member	Cpt In Definition	Date Matches	In Window	Gaps Sharing A Claim
BH_POST_DISCHARGE_7D	1248	1248	1248	1248	1248	1248	0
COPD_SPIROMETRY	1011	1011	1011	1011	1011	1011	0
DIAB_A1C_TEST	330	330	330	330	330	330	178
DIAB_BLOOD_SUGAR	531	531	531	531	531	531	178
DIAB_EYE_EXAM	486	486	486	486	486	486	0
DIAB_NEPHRO_SCREEN	498	498	498	498	498	498	0
HTN_PCP_FOLLOWUP	622	622	622	622	622	622	0

Does the check work? Known faults were injected into clean synthetic closures and the checker was run on them (recall is the share flagged; the last row is the false-positive rate on untouched rows). This tests the check, not InComm's data.

Fault	Injected	Detected	Recall
missing_claim	150	150	1.0000
wrong_member	150	150	1.0000
wrong_cpt	150	150	1.0000
shifted_date	150	150	1.0000
out_of_window	150	150	1.0000
claim_reused_across_members	150	150	1.0000
none (clean rows flagged = false positives)	2100	0	0

2. Reward attribution

- Real data: 0 of 4726 closures follow a reward case. Claims run 2025-10-29 to 2026-05-26; every case is created 2026-06-25. Temporal attribution is impossible with this export.
- Causal attribution instead (probability of necessity, Tian-Pearl bounds): 0.02 to 1.00 on the research mock. The complement is the windfall share of payouts.
- Recording offer timestamps before follow-up is the one change that makes temporal attribution possible.

3. RAF arm (separate; never summed with Stars)

A closure moves RAF only through a risk-adjustment-eligible encounter whose diagnosis maps to a payment HCC not already coded this year. Rewards must never depend on a diagnosis. Only V28 factors verified from CMS materials are used.

Care Gap Code	Ra Eligible Encounter	Payment Hcc Dx Share	Dominant Condition	Raf Value Usd Base	Raf Value Upper Bound Usd	Status
BH_POST_DISCHARGE_7D	Yes	0	none	0	n/a	ZERO: diagnosis is not a payment HCC
COPD_SPIROMETRY	No	1.0000	COPD	0	n/a	ZERO: not a risk-adjustment-eligible encounter

DIAB_A1C_TEST	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligible encounter
DIAB_BLOOD_SUGAR	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligible encounter
DIAB_EYE_EXAM	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligible encounter
DIAB_NEPHRO_SCREEN	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligible encounter
HTN_PCP_FOLLOWUP	Yes	1.0000	heart failure	n/a	4,811	NOT_AVAILABLE: V28 factor not verified (upper bound shown)